

Balance Sheet Velocity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Balance Sheet Velocity (BSV), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on BSV achieves an annualized gross (net) Sharpe ratio of 0.44 (0.39), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (25) bps/month with a t-statistic of 2.77 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes) is 19 bps/month with a t-statistic of 1.99.

1 Introduction

Asset prices reflect investors’ marginal rates of substitution between consumption today and consumption tomorrow, yet the precise mechanisms through which firm characteristics relate to systematic consumption risk remain incompletely understood. While classical consumption-based models establish that expected returns compensate investors for bearing covariance with aggregate consumption growth, empirical work continues to uncover cross-sectional patterns that challenge our understanding of which firm characteristics proxy for exposure to consumption risk. We identify Balance Sheet Velocity (BSV) as a novel predictor of stock returns that captures firms’ sensitivity to intertemporal consumption dynamics. BSV measures the rate at which firms convert balance sheet resources into operational outcomes, reflecting their ability to adapt to changing economic conditions that affect households’ consumption-investment decisions. This characteristic potentially exposes investors to systematic consumption risk during periods when marginal utility is high, particularly during economic downturns when consumption smoothing becomes more valuable.

Balance Sheet Velocity captures firms’ exposure to long-run consumption risk through multiple economic channels. First, firms with high BSV operate with greater operational flexibility, allowing them to adjust rapidly to consumption shocks that propagate through the economy ([Bansal and Yaron, 2004](#)). These firms’ cash flows exhibit higher sensitivity to persistent components in consumption growth, exposing investors to long-run risks that command higher risk premia in equilibrium ([Hansen et al., 2008](#)). The consumption-based asset pricing framework predicts that assets whose payoffs covary positively with consumption growth when marginal utility is high—during bad economic states—should earn higher expected returns to compensate investors for bearing this systematic risk. Second, BSV relates to firms’ exposure to rare disaster risk, as high-velocity firms depend more heavily on continuous eco-

nomic activity and suffer disproportionately during consumption disasters (Barro, 2006). During economic contractions when households face binding constraints on consumption smoothing, high-BSV firms experience sharper declines in profitability because their business models require sustained transaction volumes. This state-dependent exposure amplifies their systematic risk precisely when the representative agent’s marginal utility peaks. Third, the velocity metric captures firms’ sensitivity to habit formation in consumption patterns (Campbell and Cochrane, 1999). High-BSV firms typically serve markets where demand closely tracks the surplus consumption ratio—the distance between consumption and habit levels. When consumption approaches habit levels during economic downturns, these firms face severe demand contractions that translate into lower cash flows. The covariance between their returns and the intertemporal marginal rate of substitution therefore increases, necessitating higher equilibrium returns. These mechanisms collectively suggest that BSV proxies for multiple dimensions of consumption risk exposure, from long-run persistence to disaster sensitivity.

We find that a value-weighted long/short trading strategy based on Balance Sheet Velocity achieves an annualized gross Sharpe ratio of 0.44, with the high-BSV portfolio earning 26 basis points per month more than the low-BSV portfolio (t-statistic = 2.68). The strategy’s alpha remains economically and statistically significant across standard factor models, with the six-factor alpha reaching 26 basis points per month (t-statistic = 2.77). These results demonstrate that BSV captures a dimension of systematic risk not explained by existing factors related to market, size, value, profitability, investment, or momentum. The BSV premium manifests strongly among large-cap stocks, where the long/short strategy within the largest size quintile generates average returns of 32 basis points per month (t-statistic = 2.60). This finding supports the consumption risk interpretation, as larger firms’ returns should better reflect systematic consumption exposures rather than idiosyncratic factors or

market frictions. The strategy’s six-factor alpha among large stocks reaches 37 basis points per month (t-statistic = 2.99), confirming that the BSV effect does not merely reflect a small-stock phenomenon. After accounting for transaction costs using high-frequency effective spread measures, the strategy maintains a net Sharpe ratio of 0.39, placing it in the 95th percentile among 212 documented anomalies. The net six-factor generalized alpha equals 25 basis points per month (t-statistic = 2.64), indicating that BSV expands the achievable investment opportunity set even after incorporating realistic trading frictions. Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the BSV strategy retains an alpha of 19 basis points per month (t-statistic = 1.99), establishing its incremental contribution to cross-sectional return prediction.

Our study extends the consumption-based asset pricing literature by identifying Balance Sheet Velocity as a characteristic that captures firms’ differential exposure to consumption risk. While (Lettau and Ludvigson, 2001) demonstrate that consumption-wealth ratios predict aggregate returns and (Parker and Julliard, 2005) show that consumption risk varies across business cycles, we establish that BSV proxies for cross-sectional differences in firms’ consumption risk exposures. Our findings complement (Yogo, 2006), who links durable consumption growth to the cross-section through firms’ product characteristics, by showing that operational velocity metrics capture broader consumption risk dimensions including long-run risks and disaster sensitivity. The paper contributes methodologically by demonstrating that balance sheet dynamics contain information about systematic risk exposures not captured by traditional accounting-based measures examined in (Fama and French, 2006) or (Novy-Marx, 2013). Unlike static leverage or profitability ratios that measure levels, BSV captures the dynamic transformation of resources that determines firms’ sensitivity to consumption fluctuations. This velocity dimension proves orthogonal to existing anomalies, as evidenced by our spanning tests against 212 factors includ-

ing those documented in recent comprehensive studies. Our results have broader implications for understanding the consumption-risk origins of cross-sectional return patterns. The strong BSV premium among large stocks, where limits to arbitrage are minimal, supports rational risk explanations over behavioral or friction-based theories. The persistence of the premium after controlling for momentum and other factors suggests that BSV captures a fundamental aspect of how firms’ operational characteristics translate into systematic risk. These findings advance our understanding of which firm characteristics serve as proxies for consumption risk exposure, helping bridge the gap between theoretical consumption-based models and empirical cross-sectional patterns in expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial statement items that enable consistent measurement across firms and over time. Our sample includes all non-financial firms with available data necessary to construct the Balance Sheet Velocity signal, spanning several decades of annual observations. Balance Sheet Velocity signal relies on two key accounting variables from firms’ financial statements. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not captured through other specific cash flow categories in the statement of cash flows. This item reflects adjustments and reclassifications that affect the balance sheet but are not classified within operating, investing, or financing activities. Liabilities (LT) measures the firm’s total liabilities as reported on the balance sheet, representing all financial obligations and claims against the firm’s assets by creditors and other parties. We construct the Balance Sheet Velocity signal by dividing assets and liabilities

other net change (AOLOCH) by total liabilities (LT). This ratio captures the velocity of balance sheet adjustments relative to the firm’s overall liability base, providing insight into the magnitude of accounting reclassifications and other balance sheet changes scaled by the firm’s financial obligations. We calculate this measure annually using fiscal year-end values and require non-missing values for both AOLOCH and LT to ensure signal reliability. To mitigate the influence of outliers, we exclude observations where total liabilities equal zero and winsorize the resulting ratio at the 1st and 99th percentiles. The Balance Sheet Velocity signal thus measures the relative intensity of balance sheet transformations that occur outside standard operating, investing, and financing classifications, normalized by the firm’s liability structure.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BSV signal. Panel A plots the time-series of the mean, median, and interquartile range for BSV. On average, the cross-sectional mean (median) BSV is -0.01 (-0.00) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input BSV data. The signal’s interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the BSV signal for the CRSP universe. On average, the BSV signal is available for 7.37% of CRSP names, which on average make up 7.93% of total market capitalization.

4 Does BSV predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BSV using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BSV portfolio and sells the low BSV portfolio. The rest

of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short BSV strategy earns an average return of 0.26% per month with a t-statistic of 2.68. The annualized Sharpe ratio of the strategy is 0.44. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.29 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 4.91 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 635 stocks and an average market capitalization of at least \$2,550 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each

portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.15. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -4-32bps/month. The lowest return, (-4 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.52. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BSV trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BSV strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BSV, as well as average returns and alphas for long/short trading BSV strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the BSV strategy achieves an average return of 32 bps/month with a t-statistic of 2.60. Among these large cap stocks, the alphas for the BSV strategy relative to the five most common factor models range from 28 to 43 bps/month

with t-statistics between 2.29 and 3.51.

5 How does BSV perform relative to the zoo?

Figure 2 puts the performance of BSV in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the BSV strategy falls in the distribution. The BSV strategy’s gross (net) Sharpe ratio of 0.44 (0.39) is greater than 84% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BSV strategy (red line).² Ignoring trading costs, a \$1 invested in the BSV strategy would have yielded \$1.79 which ranks the BSV strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BSV strategy would have yielded \$1.46 which ranks the BSV strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BSV relative to those. Panel A shows that the BSV strategy gross alphas fall between the 51 and 77 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BSV strategy has a positive net generalized alpha for five out of the five factor models. In these cases BSV ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BSV add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BSV with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BSV or at least to weaken the power BSV has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV}BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BSV and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BSV signal in these Fama-MacBeth regressions exceed 1.75, with the minimum t-statistic occurring when controlling for Advertising Expense. Controlling for all six closely related anomalies, the t-statistic on BSV is 2.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the BSV strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BSV strategy earns alphas that range from 20-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.09, which is achieved when controlling for Advertising Expense. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BSV trading strategy achieves an alpha of 19bps/month with a t-statistic of 1.99.

7 Does BSV add relative to the whole zoo?

Finally, we can ask how much adding BSV to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the BSV signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes BSV grows to \$34.13.

8 Conclusion

This study provides compelling evidence that Balance Sheet Velocity (BSV) represents a statistically and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BSV-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs. The value-weighted long/short portfolio achieves an annualized net Sharpe ratio of 0.39, with monthly abnormal returns of 25 basis points relative to the Fama-French five-factor model plus momentum, maintaining statistical significance with a t-statistic of

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BSV is available.

2.64. Notably, BSV retains predictive power even when controlling for six closely related factors from the factor zoo, generating a monthly alpha of 19 basis points with marginal statistical significance (t-statistic of 1.99). These findings suggest that BSV captures unique information about firm fundamentals not fully explained by existing factor models or related accounting-based signals. From a practical perspective, the signal’s robustness to transaction costs and its performance after controlling for related factors indicate potential implementation value for quantitative investment strategies. The economic magnitude of the returns, combined with reasonable Sharpe ratios, suggests that BSV could serve as a valuable addition to multi-factor portfolios. However, several limitations warrant consideration. The marginal statistical significance when controlling for related factors (t-statistic of 1.99) suggests some overlap with existing signals, potentially limiting its incremental contribution in highly sophisticated factor models. Additionally, while our analysis follows established protocols, the true out-of-sample performance of BSV remains to be tested as market participants increasingly incorporate such signals. Future research should explore several promising directions. First, investigating the economic mechanisms underlying BSV’s predictive power could enhance our understanding of why balance sheet efficiency translates into superior returns. Second, examining BSV’s performance across different market regimes, sectors, and international markets would establish its generalizability. Third, analyzing the signal’s decay rate and optimal rebalancing frequency could improve implementation strategies. Finally, exploring potential enhancements through machine learning techniques or combinations with other fundamental signals may unlock additional predictive power. Despite these areas for future investigation, our results establish BSV as a robust addition to the cross-sectional return prediction literature, offering both theoretical insights into the relationship between balance sheet efficiency and asset prices, and practical value for quantitative investment applications.

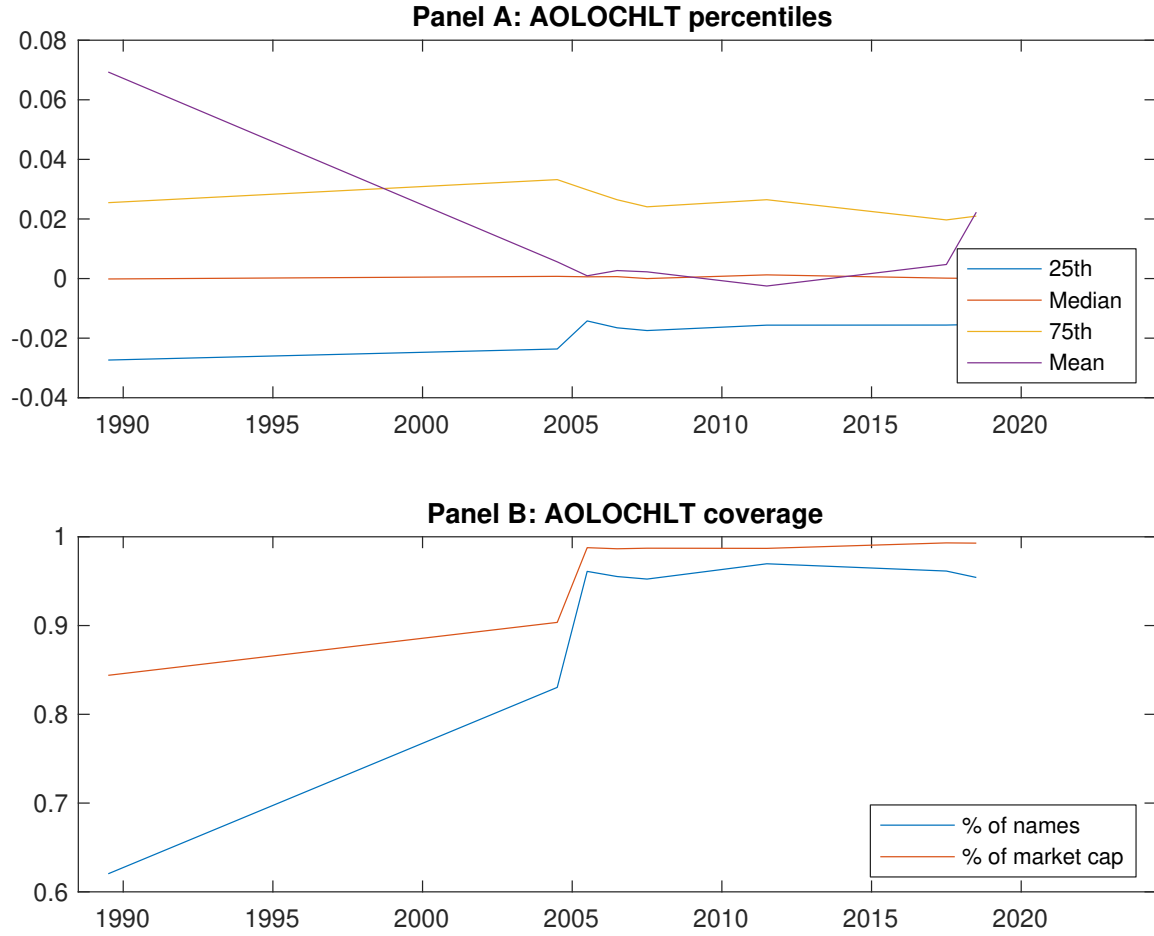


Figure 1: Times series of BSV percentiles and coverage.
This figure plots descriptive statistics for BSV. Panel A shows cross-sectional percentiles of BSV over the sample. Panel B plots the monthly coverage of BSV relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BSV. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on BSV-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.60 [2.58]	0.76 [3.61]	0.68 [3.47]	0.73 [3.50]	0.86 [3.68]	0.26 [2.68]
α_{CAPM}	-0.19 [-2.91]	0.05 [0.76]	0.03 [0.38]	0.02 [0.32]	0.07 [1.01]	0.26 [2.68]
α_{FF3}	-0.17 [-2.67]	0.01 [0.26]	-0.02 [-0.37]	-0.01 [-0.17]	0.12 [2.06]	0.29 [3.05]
α_{FF4}	-0.11 [-1.78]	0.03 [0.49]	-0.01 [-0.10]	-0.01 [-0.18]	0.11 [1.78]	0.22 [2.29]
α_{FF5}	-0.11 [-1.74]	-0.05 [-0.85]	-0.12 [-2.11]	-0.06 [-1.23]	0.22 [3.80]	0.33 [3.41]
α_{FF6}	-0.06 [-1.03]	-0.03 [-0.56]	-0.10 [-1.78]	-0.06 [-1.16]	0.20 [3.44]	0.26 [2.77]
Panel B: Fama and French (2018) 6-factor model loadings for BSV-sorted portfolios						
β_{MKT}	1.01 [64.51]	1.01 [68.91]	0.94 [66.78]	1.00 [75.86]	1.01 [69.41]	-0.00 [-0.06]
β_{SMB}	0.08 [3.43]	-0.02 [-0.88]	-0.01 [-0.39]	-0.06 [-3.36]	-0.01 [-0.40]	-0.08 [-2.49]
β_{HML}	-0.09 [-3.32]	0.06 [2.50]	0.12 [4.92]	0.06 [2.43]	-0.12 [-4.79]	-0.03 [-0.74]
β_{RMW}	-0.11 [-3.88]	0.08 [3.04]	0.18 [6.92]	0.09 [3.81]	-0.14 [-5.11]	-0.02 [-0.57]
β_{CMA}	0.01 [0.22]	0.14 [3.83]	0.11 [2.95]	0.07 [2.07]	-0.20 [-5.34]	-0.21 [-3.39]
β_{UMD}	-0.07 [-5.25]	-0.03 [-2.01]	-0.03 [-2.22]	-0.00 [-0.39]	0.03 [2.42]	0.10 [4.91]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1070	644	682	635	1031	
me (\$10 ⁶)	2550	2979	3241	3598	3980	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the BSV strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [2.68]	0.26 [2.68]	0.29 [3.05]	0.22 [2.29]	0.33 [3.41]	0.26 [2.77]
Quintile	NYSE	EW	0.19 [3.15]	0.20 [3.39]	0.20 [3.44]	0.14 [2.45]	0.20 [3.30]	0.15 [2.50]
Quintile	Name	VW	0.29 [2.41]	0.30 [2.47]	0.32 [2.79]	0.25 [2.14]	0.36 [3.02]	0.29 [2.46]
Quintile	Cap	VW	0.30 [2.88]	0.28 [2.65]	0.33 [3.31]	0.26 [2.58]	0.40 [3.99]	0.33 [3.35]
Decile	NYSE	VW	0.35 [2.45]	0.35 [2.39]	0.40 [2.85]	0.29 [2.12]	0.47 [3.29]	0.37 [2.66]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.38]	0.23 [2.34]	0.25 [2.66]	0.21 [2.22]	0.29 [3.02]	0.25 [2.64]
Quintile	NYSE	EW	-0.04 [-0.52]					
Quintile	Name	VW	0.25 [2.13]	0.25 [2.10]	0.28 [2.36]	0.23 [1.97]	0.31 [2.60]	0.27 [2.25]
Quintile	Cap	VW	0.28 [2.63]	0.27 [2.47]	0.31 [3.07]	0.26 [2.65]	0.37 [3.74]	0.33 [3.36]
Decile	NYSE	VW	0.32 [2.19]	0.30 [2.00]	0.33 [2.38]	0.27 [1.96]	0.40 [2.83]	0.34 [2.45]

Table 3: Conditional sort on size and BSV

This table presents results for conditional double sorts on size and BSV. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BSV. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BSV and short stocks with low BSV. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	BSV Quintiles					BSV Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.50 [1.43]	0.69 [2.13]	0.73 [2.48]	0.80 [2.64]	0.68 [1.93]	0.18 [1.74]	0.17 [1.62]	0.16 [1.52]	0.11 [1.01]	0.18 [1.68]	0.14 [1.26]
	(2)	0.67 [2.02]	0.68 [2.40]	0.82 [3.01]	0.82 [2.90]	0.67 [2.14]	0.00 [0.00]	0.07 [0.64]	0.07 [0.67]	0.06 [0.58]	0.05 [0.45]	0.04 [0.37]
	(3)	0.76 [2.55]	0.64 [2.47]	0.75 [2.93]	0.81 [3.15]	0.81 [2.76]	0.05 [0.52]	0.05 [0.48]	0.06 [0.61]	0.06 [0.57]	0.15 [1.40]	0.14 [1.27]
	(4)	0.73 [2.77]	0.79 [3.29]	0.72 [3.07]	0.84 [3.62]	0.89 [3.20]	0.16 [1.53]	0.14 [1.28]	0.20 [2.03]	0.13 [1.33]	0.33 [3.42]	0.27 [2.77]
	(5)	0.57 [2.53]	0.77 [3.76]	0.70 [3.56]	0.75 [3.58]	0.89 [3.73]	0.32 [2.60]	0.30 [2.34]	0.35 [2.86]	0.28 [2.29]	0.43 [3.51]	0.37 [2.99]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	BSV Quintiles					BSV Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	451	451	450	449	449	47	55	55	54	51	
	(2)	133	133	132	133	133	91	93	94	95	93	
	(3)	90	90	90	90	90	158	159	159	162	160	
	(4)	74	74	74	74	74	326	342	341	343	341	
(5)	66	66	66	66	66	2612	2266	2430	2939	2882		

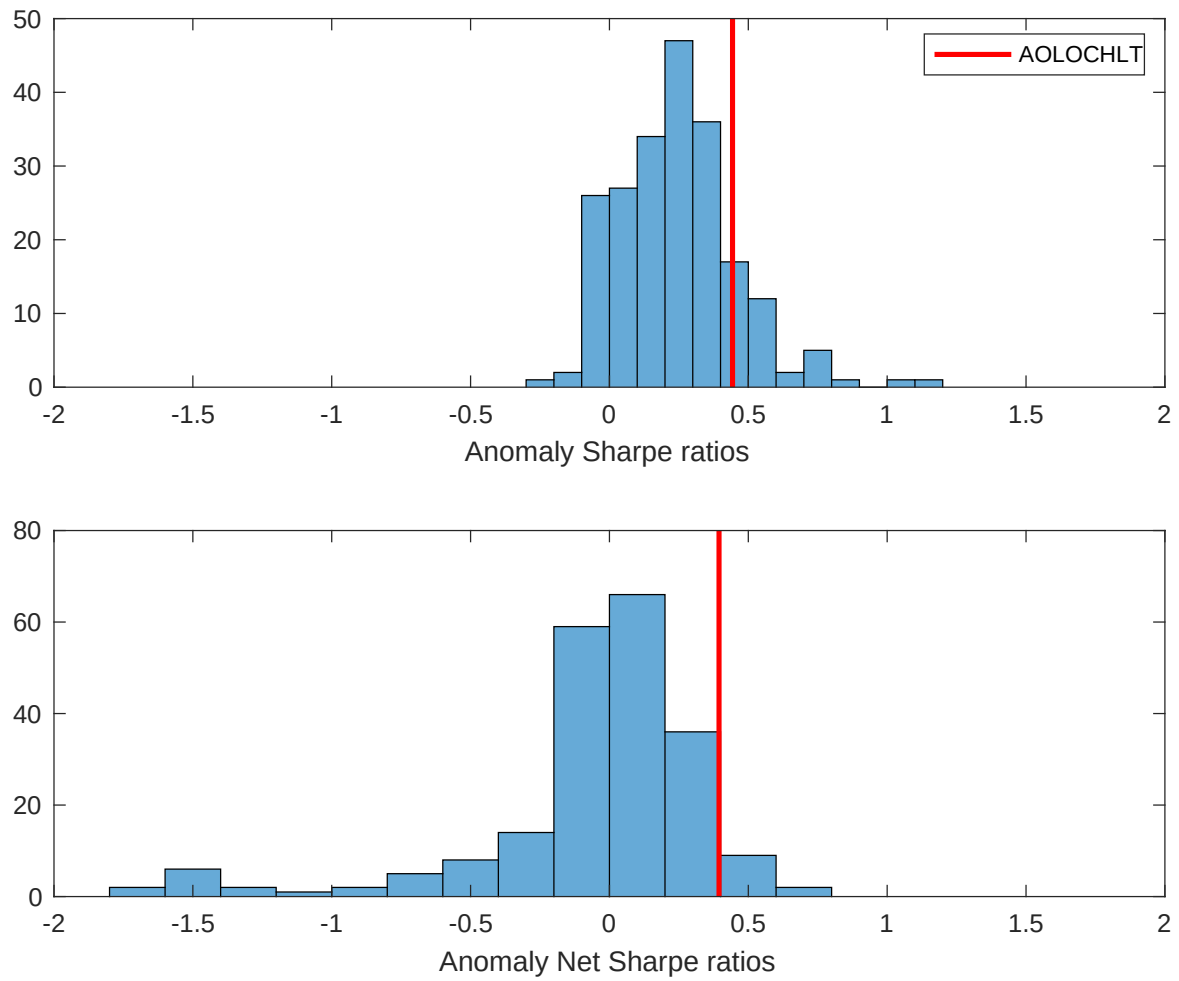


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BSV with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

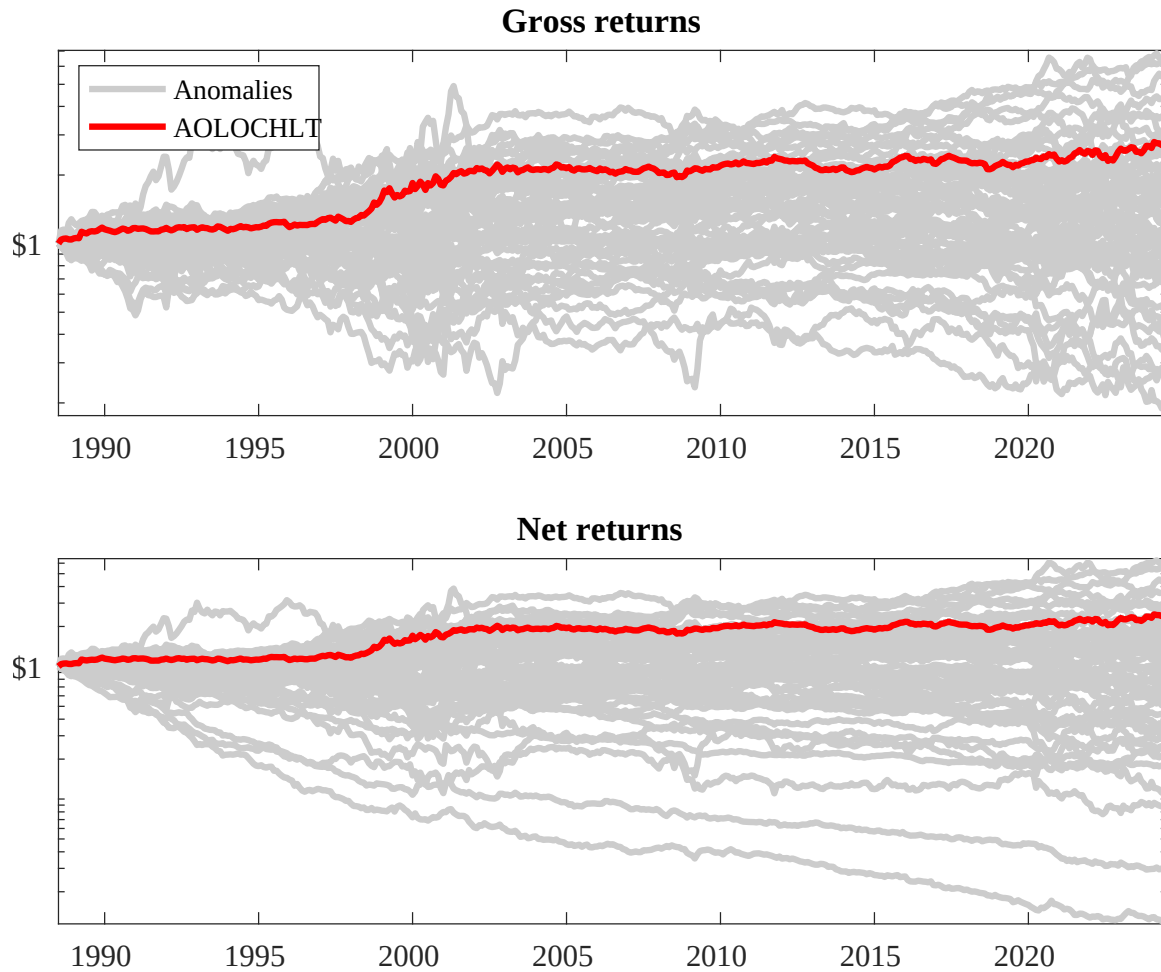
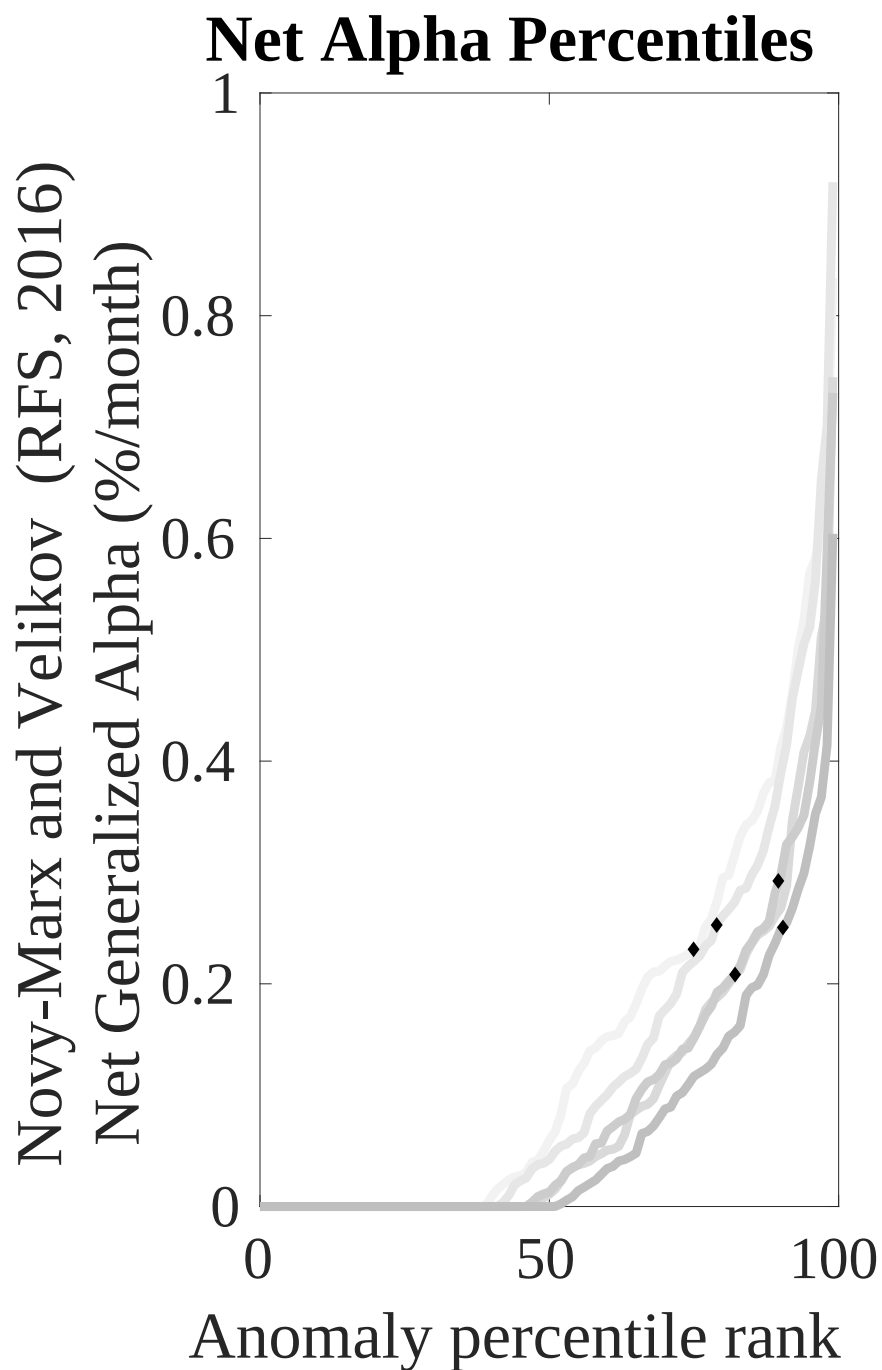
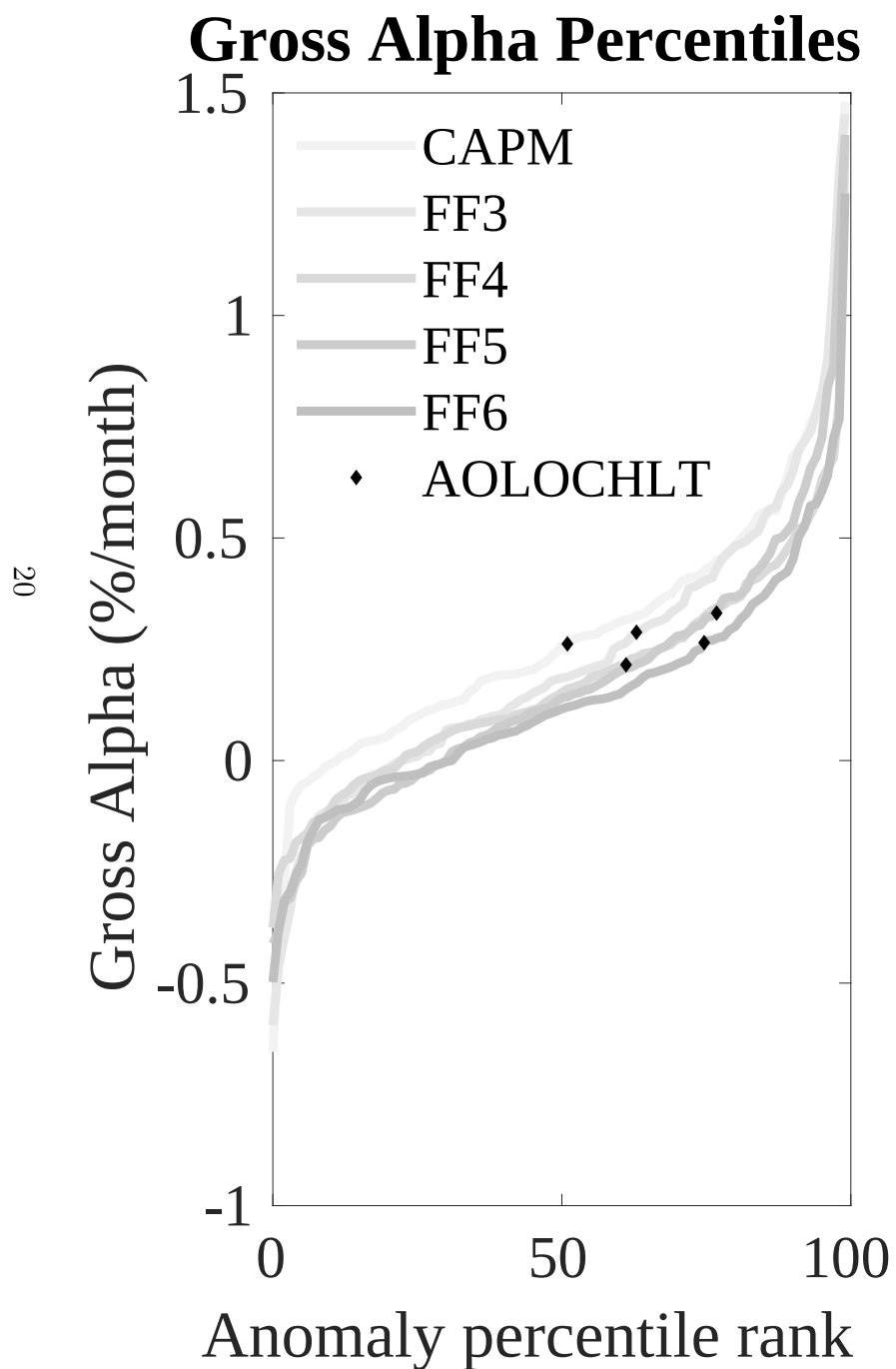


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BSV trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



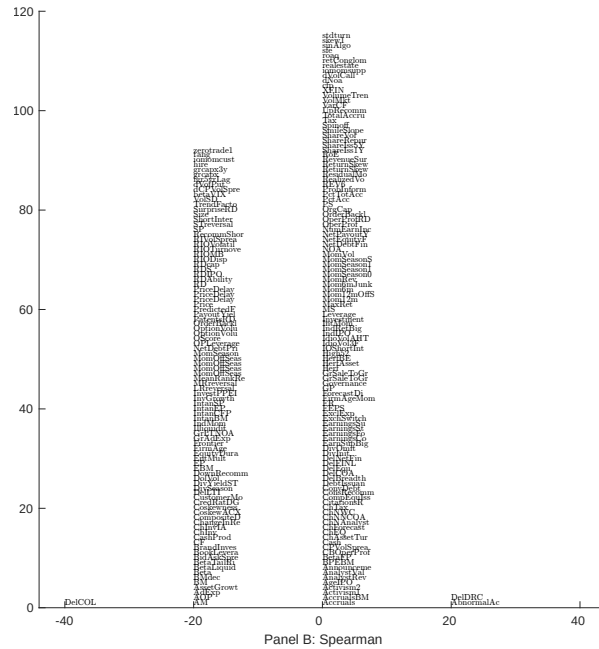
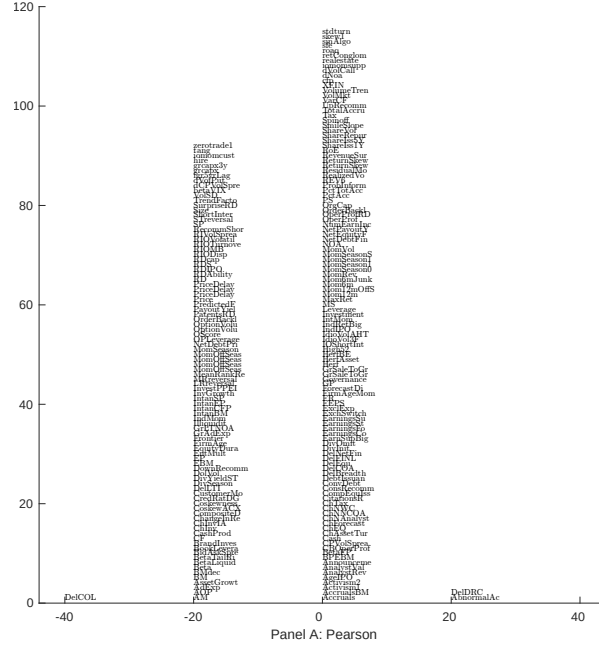


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with BSV. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

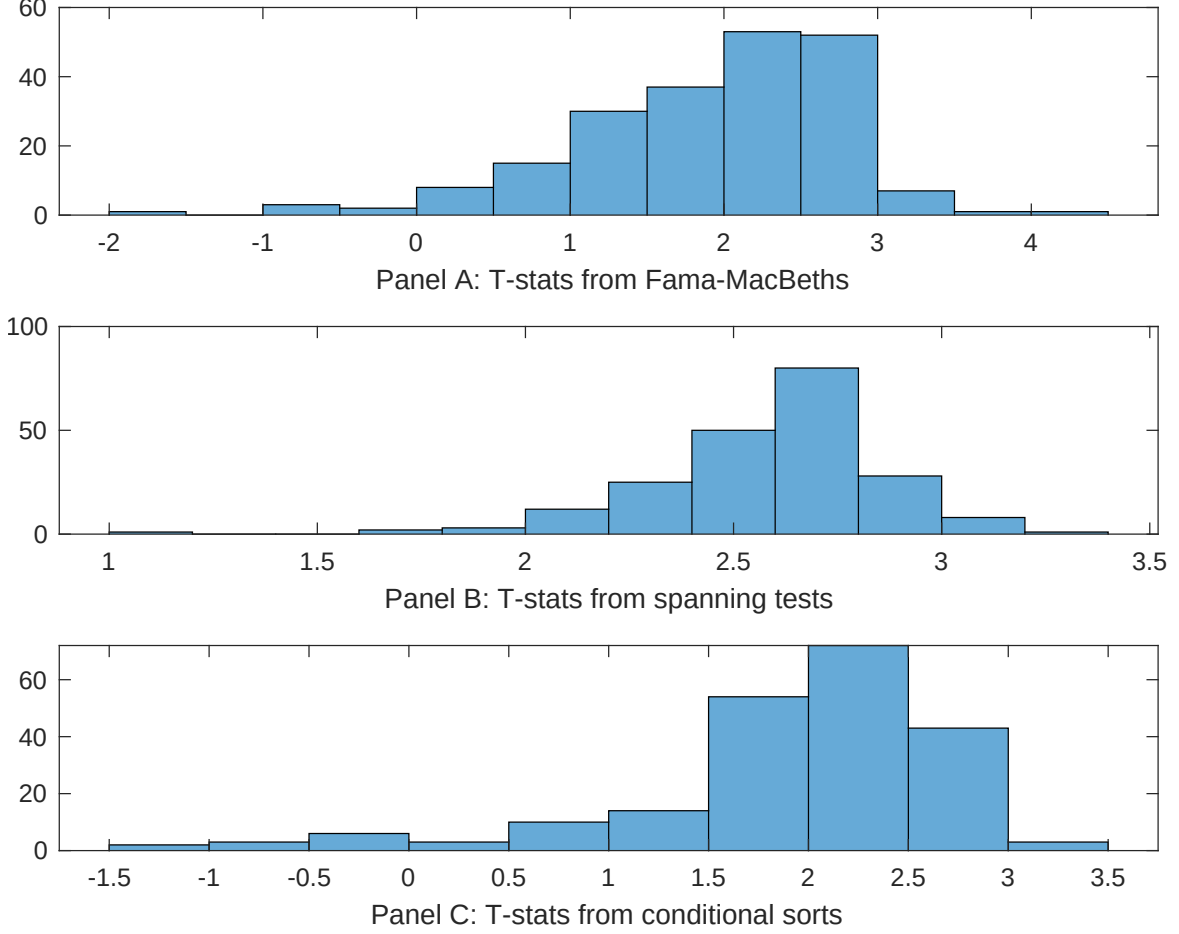


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV}BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BSV. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BSV}BSV_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11 [3.99]	0.11 [3.58]	0.11 [3.88]	0.94 [3.32]	0.11 [3.88]	0.11 [3.91]	0.12 [4.22]
BSV	0.64 [3.11]	0.51 [2.57]	0.40 [1.89]	0.52 [1.75]	0.45 [2.10]	0.39 [2.05]	0.71 [2.43]
Anomaly 1	0.21 [4.69]						0.24 [3.38]
Anomaly 2		0.14 [1.45]					0.19 [1.72]
Anomaly 3			0.51 [1.90]				0.76 [2.46]
Anomaly 4				0.56 [0.08]			-0.14 [-0.21]
Anomaly 5					0.56 [0.19]		-0.49 [-1.35]
Anomaly 6						0.14 [5.62]	0.14 [4.73]
# months	432	432	432	427	432	432	427
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BSV trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BSV} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.20 [2.09]	0.24 [2.49]	0.28 [2.91]	0.28 [2.83]	0.22 [2.24]	0.27 [2.83]	0.19 [1.99]
Anomaly 1	-22.95 [-5.21]						-12.80 [-2.50]
Anomaly 2		15.56 [4.46]					6.82 [1.74]
Anomaly 3			-15.31 [-3.46]				-5.20 [-1.07]
Anomaly 4				-2.61 [-0.88]			0.58 [0.20]
Anomaly 5					15.74 [3.52]		10.15 [2.26]
Anomaly 6						17.60 [4.57]	12.09 [3.06]
mkt	0.29 [0.12]	1.02 [0.43]	0.09 [0.04]	-0.26 [-0.11]	-1.90 [-0.79]	-1.93 [-0.81]	-1.90 [-0.79]
smb	-10.32 [-3.05]	-11.54 [-3.33]	-9.35 [-2.73]	-7.09 [-1.91]	-9.22 [-2.69]	-8.26 [-2.45]	-12.30 [-3.39]
hml	8.20 [1.77]	8.89 [1.81]	-4.51 [-1.08]	-1.94 [-0.44]	-0.98 [-0.23]	-0.48 [-0.12]	10.84 [2.04]
rmw	-1.87 [-0.44]	1.26 [0.29]	-5.69 [-1.29]	-1.85 [-0.41]	-0.38 [-0.09]	-2.59 [-0.61]	-0.74 [-0.17]
cma	-9.12 [-1.43]	-20.72 [-3.45]	-10.57 [-1.56]	-20.14 [-3.19]	-17.61 [-2.87]	-19.06 [-3.17]	-7.15 [-1.01]
umd	9.24 [4.42]	9.87 [4.70]	9.06 [4.22]	9.29 [3.82]	9.67 [4.55]	5.83 [2.52]	5.65 [2.30]
# months	432	432	432	428	432	432	428
$\bar{R}^2(\%)$	18	17	15	13	15	17	22

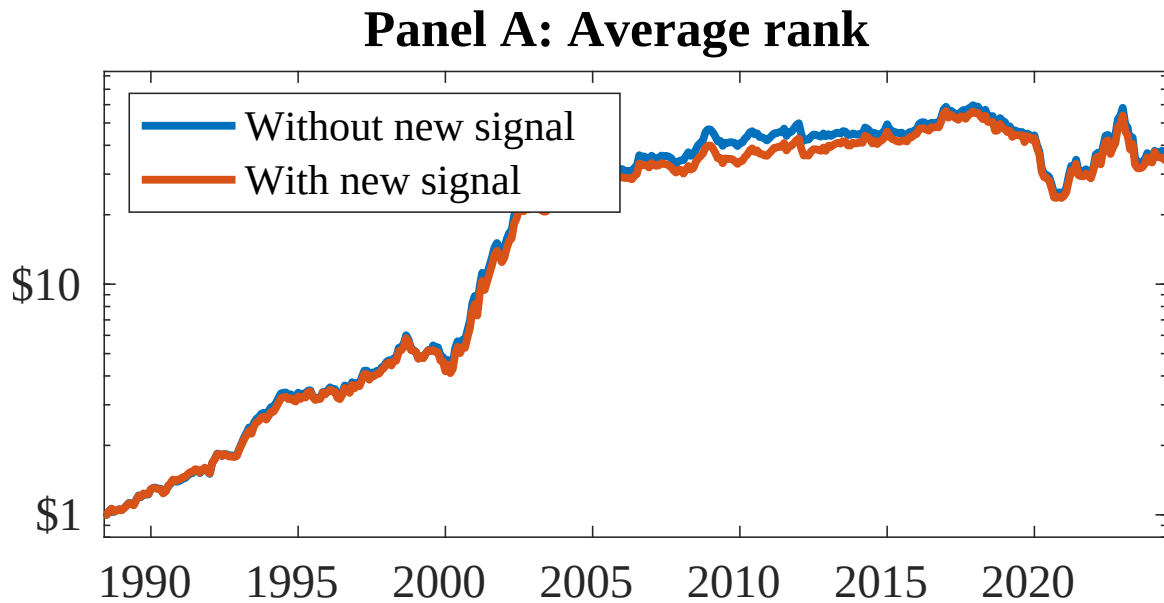


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as BSV. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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